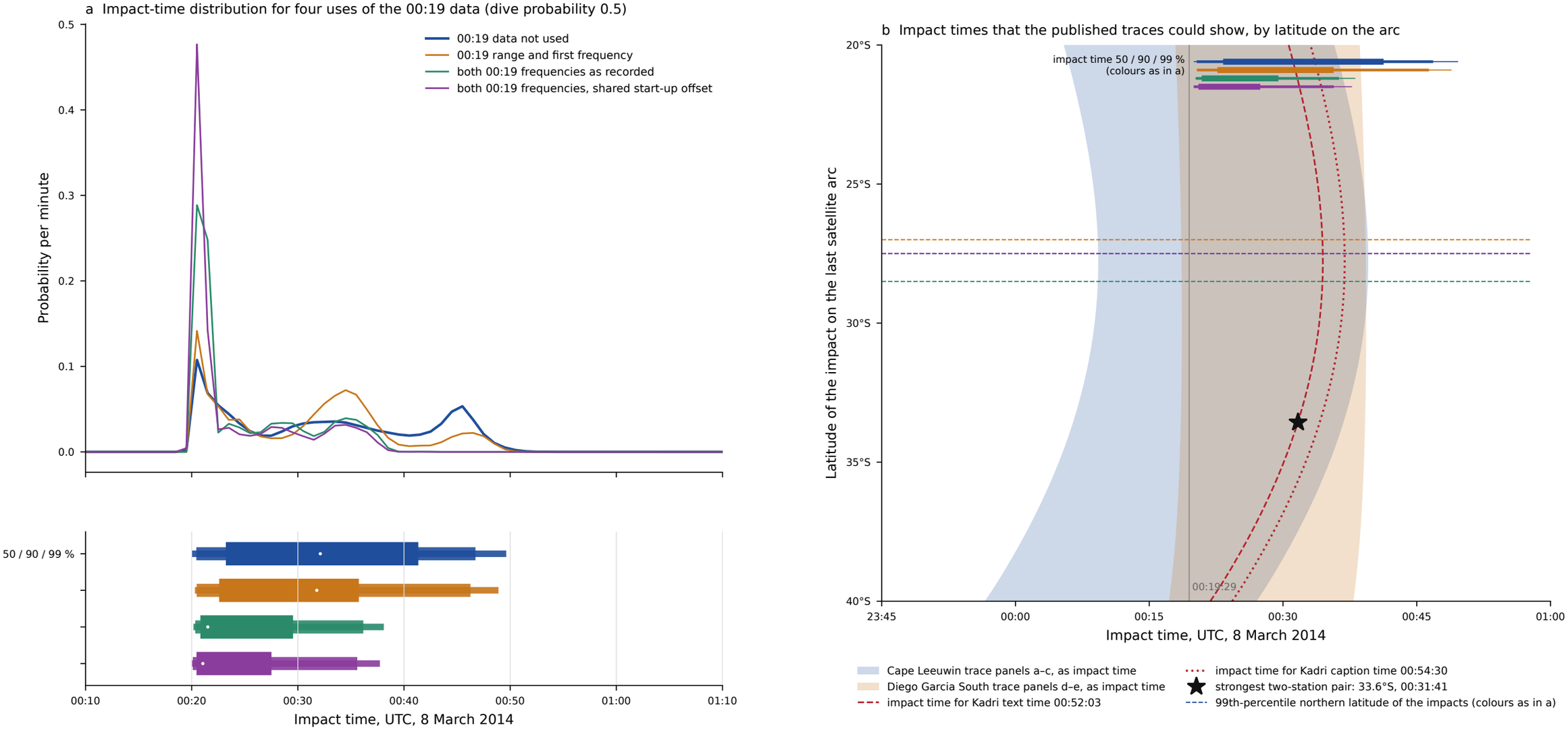
When the aircraft hit the water, against the impact times that the published Cape Leeuwin and Diego Garcia traces could show



Plain note: Panel a shows when the aircraft hit the water, for four ways to use the last satellite data at 00:19. The bars below the curves give the 50 %, 90 % and 99 % ranges. Panel b shows, for each place along the last satellite arc, which impact times the published Cape Leeuwin and Diego Garcia traces could show. The red lines give the impact time that each of Kadri's two signal times would need. The star is the place and time that the strongest pair of candidate signals would need. Where the shaded bands do not cover the impact times, the published traces could not show the impact even if it were loud enough.

Technical note: Source posterior: end-of-flight run J + northern supplement, p = 0.5 (run_j_windows.py, prereg bc99b07); densities = pooled 2 s histograms of the strata mixed with the supplement by its share (D0′ 0.125), shown per minute; bars = [q25, q75], [q5, q95], [q0.5, q99.5] of windows.csv, dot = median. Arc = offset-0 line of the block-2 TL grid (table_points.json). Bands in b = panel span minus geodesic / c, c = 1.482 +/- 2 x 0.006 km/s; Kadri 2024 Fig. 9 panels a–c 00:27–00:57 (H01W), d–e 01:00–01:20 (H08S), digitised. Red lines: impact time = H01W arrival − range / 1.482 km/s for the text time 00:52:03 and the caption time 00:54:30. Star: locus of the strongest two-station pair (lag 1,245 s, 33.56 S 94.51 E, implied impact 00:31:41; composition note); its bearing from H01W is 268.9 deg, not Kadri's 57 or 306.18 deg (nearest Table 1 bearing 00:52:50, 274.84 deg). Dashed horizontal lines: 99th-percentile northern latitude of the impact mass per option. Latitude span to 20 S. ESS: D0′ (held out) 1,336,700, D2 (R600 BTO + BFO) 107,226, D3 (both BFOs raw) 5,671, D4 (both BFOs + shared start-up offset) 25,414. PREDICTIVE, NOT CALIBRATED. C1: propagation slope misfit +10.4 / +7.4 dB per octave at H01W / H08S (after the EW0112 source result a source-term caveat; air9 verdict NOT EXPLAINED, gap G23). C2: coupling efficiency eta not calibrated; swept over four decades; F-35A upper anchor eta = 2.1e-4 is provisional (from Brown 2026's quote of Metz 2023; re-derivation from Metz pending). No detection is claimed.